

Repository Dependencies

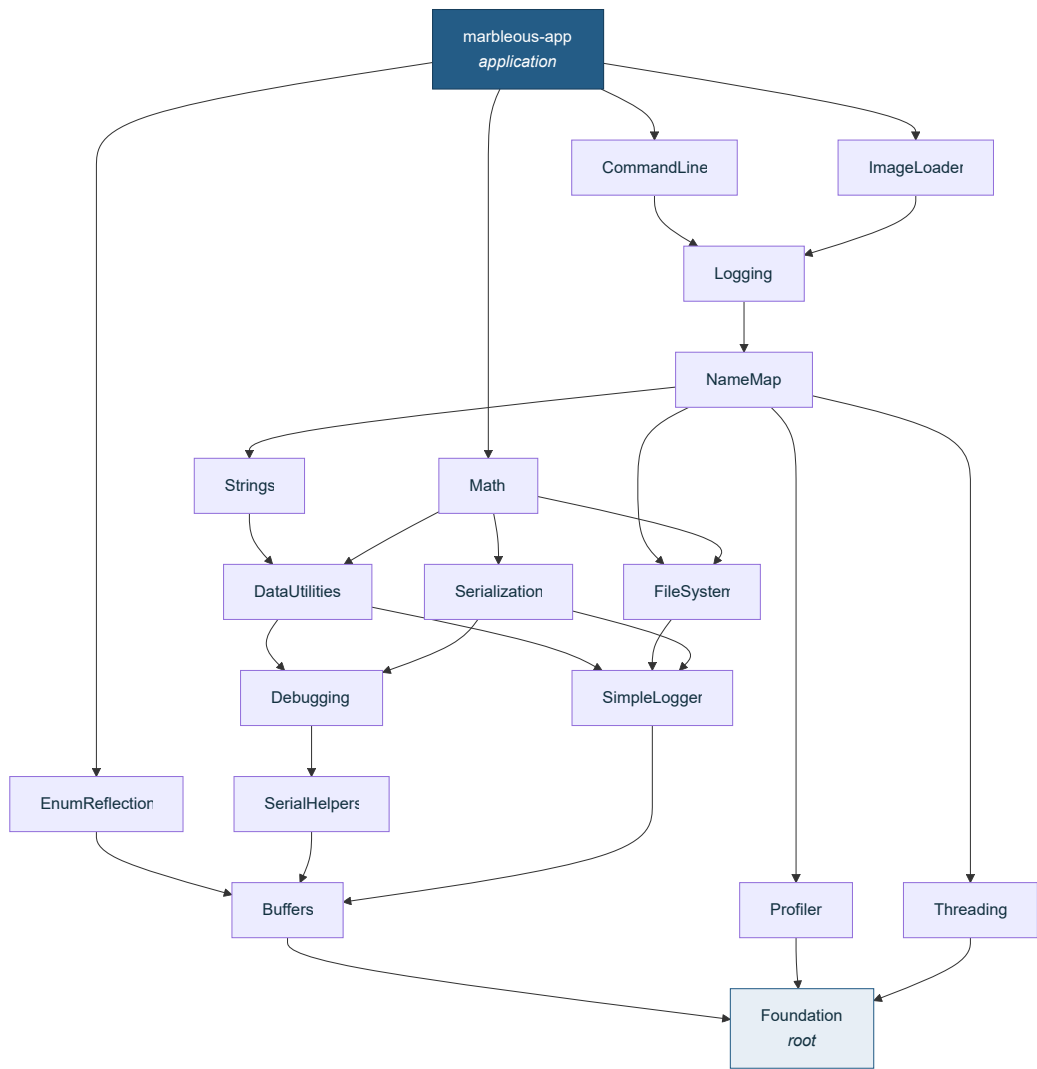
Generated from REPO_CONFIGS in scripts/project_utils.py by scripts/generate_dependency_graph.py . Do not edit by hand.

- **Repositories:** 18
- **Dependency edges:** 71
- **Cycles:** 0

The three vendored trees — glad , glfw and external — are omitted: they have no ProjectConfig and are reached by raw include path rather than through the dependency mechanism.

Graph

Shown as the transitive reduction: 44 of the 71 declared edges are implied by others, leaving 27 that carry structure. Arrows point from dependent to dependency.



Layers

The graph is acyclic, so it has a strict total ordering. A repo at any level depends only on levels beneath it.

- **L0** — Foundation
- **L1** — Buffers
- **L2** — EnumReflection
- **L3** — SerialHelpers
- **L4** — Debugging
- **L5** — SimpleLogger
- **L6** — DataUtilities
- **L7** — FileSystem
- **L8** — Profiler
- **L9** — Strings

- **L10** — Threading
- **L11** — NameMap
- **L12** — Logging
- **L13** — Serialization
- **L14** — Math
- **L15** — CommandLine
- **L16** — ImageLoader
- **L17** — marbleous-app

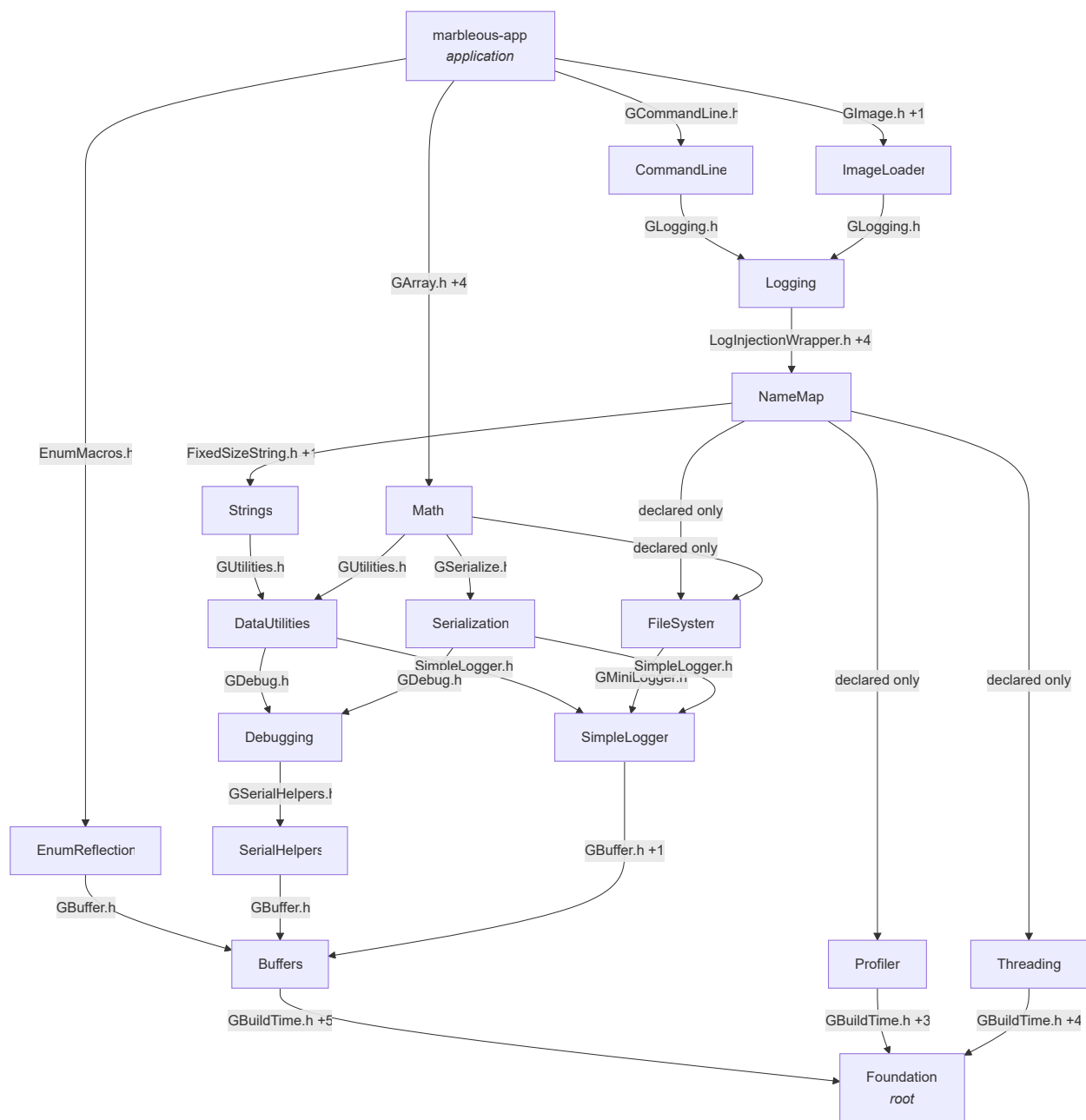
Cycles

None. Every repository can be built, versioned and published on its own.

Include graph

Repository dependencies exist because of `#include` directives. The 68 cross-repo edges below are backed by 155 distinct repo/header references, scanned from the generated repositories.

The same spine as above, with each edge labelled by the header that creates it:



Every cross-repo edge

`test` marks an edge that exists only because of test code — the library itself does not need it.

From	To	Headers	Origin
Buffers	Foundation	GBuildTime.h, GMacros.h, GPortable.h, GPrintfHelpers.h, GTypes.h, Warnings.h	lib + test
CommandLine	DataUtilities	GString.h	lib
CommandLine	Debugging	GDebug.h	lib
CommandLine	Foundation	GBuildTime.h, GPortable.h, GTypes.h, Warnings.h	lib + test
CommandLine	Logging	GLogging.h	lib
DataUtilities	Buffers	GBuffer.h	lib
DataUtilities	Debugging	GDebug.h	lib
DataUtilities	Foundation	GBuildTime.h, GMacros.h, GPortable.h, Warnings.h	lib
DataUtilities	SerialHelpers	GSerialHelpers.h	lib
DataUtilities	SimpleLogger	SimpleLogger.h	lib
Debugging	Buffers	GBuffer.h, GUnicode.h	lib + test
Debugging	Foundation	GBuildTime.h, GEmbeddedData.h, GMacros.h, GPortable.h, GTypes.h, Warnings.h	lib + test
Debugging	SerialHelpers	GSerialHelpers.h	lib
EnumReflection	Buffers	GBuffer.h	lib
EnumReflection	Foundation	GBuildTime.h, GMacros.h, GTypes.h, Warnings.h	lib + test
EnumReflection	Serialization	GSerialize.h	test
FileSystem	Buffers	GUnicode.h	lib
FileSystem	Foundation	GBuildTime.h, GPortable.h, GTypes.h, Warnings.h	lib + test
FileSystem	SimpleLogger	GMiniLogger.h	lib
Foundation	Buffers	GBuffer.h, GUnicode.h	test
ImageLoader	FileSystem	GFileSystem.h, ResourceHelper.h	lib
ImageLoader	Foundation	GBuildTime.h, GMacros.h, GPortable.h, GTypes.h, Warnings.h	lib + test
ImageLoader	Logging	GLogging.h	lib
ImageLoader	Threading	tinythread.h	lib
Logging	Buffers	GBuffer.h, GUnicode.h	lib
Logging	Foundation	GBuildTime.h, GMacros.h, GPortable.h, Warnings.h	lib
Logging	NameMap	LogInjectionWrapper.h, NameMap.h, NameMap.hpp, NameMapInterface.h, NodeVisitorLog.h	lib
Logging	SimpleLogger	LoggingMacros.h, SimpleLogger.h	lib
Logging	Strings	FixedSizeStringInterface.h, StringSanitizer.h	lib
Logging	Threading	RWLockInterface.h, tinythread.h	lib
Math	Buffers	GBuffer.h	lib + test
Math	DataUtilities	GUtilities.h	lib
Math	Debugging	GDebug.h, GTrace.h	lib
Math	Foundation	GBuildTime.h, GMacros.h, GPortable.h, GTypes.h, Warnings.h	lib + test
Math	SerialHelpers	GSerialHelpers.h	lib
Math	Serialization	GSerialize.h	lib
NameMap	Debugging	GDebug.h, GTestUtilities.h, GTrace.h	lib + test
NameMap	Foundation	GBuildTime.h, GEmbeddedData.h, GMacros.h, GPortable.h, GTypes.h, Warnings.h	lib + test
NameMap	Logging	GLogging.h	test
NameMap	SimpleLogger	LoggingMacros.h, SimpleLogger.h	test
NameMap	Strings	FixedSizeString.h, FixedSizeStringInterface.h	lib
Profiler	Foundation	GBuildTime.h, GMacros.h, GPortable.h, Warnings.h	lib + test
SerialHelpers	Buffers	GBuffer.h	lib
SerialHelpers	Foundation	GPortable.h, GPrintfHelpers.h	lib
Serialization	Buffers	GBuffer.h, GUnicode.h	lib + test
Serialization	Debugging	GDebug.h	lib
Serialization	Foundation	GBuildTime.h, GPortable.h, GTypes.h, Warnings.h	lib + test
Serialization	SimpleLogger	SimpleLogger.h	lib
SimpleLogger	Buffers	GBuffer.h, GUnicode.h	lib
SimpleLogger	Foundation	GBuildTime.h, GMacros.h, GPortable.h, GTypes.h, Warnings.h	lib + test
Strings	DataUtilities	GUtilities.h	lib + test
Strings	Debugging	GDebug.h, GTestUtilities.h	lib + test
Strings	Foundation	GBuildTime.h, GPortable.h, GTypes.h, Warnings.h	lib + test
Strings	SimpleLogger	SimpleLogger.h	lib
Threading	Foundation	GBuildTime.h, GMacros.h, GPortable.h, GTypes.h, Warnings.h	lib + test
marbleous-app	Buffers	GBuffer.h, GUnicode.h	lib
marbleous-app	CommandLine	GCommandLine.h	lib
marbleous-app	DataUtilities	GUtilities.h	lib
marbleous-app	Debugging	GDebug.h	lib
marbleous-app	EnumReflection	EnumMacros.h	lib
marbleous-app	Foundation	GBuildTime.h, GMacros.h, GPortable.h, GTypes.h, Variadic.h, Warnings.h	lib
marbleous-app	ImageLoader	GImage.h, ImageLoader.h	lib
marbleous-app	Logging	GLogging.h	lib

From	To	Headers	Origin
marbleous-app	Math	GArray.h, GArrayOfStructs.h, GComponentType.h, GQuaternion.h, GUnits.h	lib
marbleous-app	NameMap	NameMap.h	lib
marbleous-app	Serialization	GSerialize.h	lib
marbleous-app	SimpleLogger	SimpleLogger.h	lib
marbleous-app	Threading	RWLockAtomic.h, tinythread.h	lib

Consistency

Declared but unused (6). REPO_CONFIGS names these dependencies, but no source includes anything from them. They add include paths the compiler does not need:

- Math → FileSystem
- NameMap → DataUtilities
- NameMap → FileSystem
- NameMap → Profiler
- NameMap → SerialHelpers
- NameMap → Threading

Used but undeclared (3). These includes resolve only because the include path arrives transitively through another dependency. They compile today, but would break if that intermediate dependency were removed:

- EnumReflection → Serialization (GSerialize.h)
- Foundation → Buffers (GBuffer.h, GUnicode.h)
- NameMap → Logging (GLogging.h)

All repositories

Level	Repo	Kind	Used by	Direct	Transitive	Depends on
L0	Foundation	library	17	0	0	—
L1	Buffers	library	10	1	1	Foundation
L2	EnumReflection	header-only	1	2	2	Buffers, Foundation
L3	SerialHelpers	header-only	4	2	2	Buffers, Foundation
L4	Debugging	library	7	3	3	Buffers, Foundation, SerialHelpers
L5	SimpleLogger	library	7	2	2	Buffers, Foundation
L6	DataUtilities	library	5	5	5	Buffers, Debugging, Foundation, SerialHelpers, SimpleLogger
L7	FileSystem	library	3	3	3	Buffers, Foundation, SimpleLogger
L8	Profiler	library	1	1	1	Foundation
L9	Strings	library	2	4	6	DataUtilities, Debugging, Foundation, SimpleLogger
L10	Threading	library	4	1	1	Foundation
L11	NameMap	library	2	9	10	DataUtilities, Debugging, FileSystem, Foundation, Profiler, SerialHelpers, SimpleLogger, Strings, Threading
L12	Logging	library	3	6	11	Buffers, Foundation, NameMap, SimpleLogger, Strings, Threading
L13	Serialization	library	2	4	5	Buffers, Debugging, Foundation, SimpleLogger
L14	Math	library	1	7	8	Buffers, DataUtilities, Debugging, FileSystem, Foundation, SerialHelpers, Serialization
L15	CommandLine	library	1	4	12	DataUtilities, Debugging, Foundation, Logging
L16	ImageLoader	library	1	4	12	FileSystem, Foundation, Logging, Threading
L17	marbleous-app	application	0	13	17	Buffers, CommandLine, DataUtilities, Debugging, EnumReflection, Foundation, ImageLoader, Logging, Math, NameMap, Serialization, SimpleLogger, Threading